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Life-cycle model of terrestrial carbon exchange

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ABSTRACT

We present here a terrestrial carbon cycle model based on a scheme of the phytomass change, which is continuous in time. The experimental information about net primary production, living and dead phytomass, and soil organic matter for various ecosystems is used for calibration of the model. The suggested model enables to characterize terrestrial ecosystems as carbon sources or carbon sinks and to evaluate intensity of these sources and sinks. The model is applied for the European territory of Russia as a case study. Intensity of the total exchange carbon flux for this territory is evaluated. The obtained results allow to conclude that the given territory is the sink of carbon.

1. Introduction

Today, when human activities have an increasing impact on the biosphere, to find a balanced evolution between the anthroposphere and environment is the fundamental problem. The carbon cycle is an important component in study of climate change impact on natural vegetation and vegetation feedback on climate. At present, there are a lot of different models of the carbon cycle both for the global level and the local ecosystems. Basic structures of these models are more or less similar: they consist of three compartments (atmosphere, living, and dead organic matter) and three flows (production of organic matter, litter fall, and decomposition of dead organic matter). The carbon cycle models differ from each other by their complexity and details of parameterizations (subdividing the compartments and expanding the flows).

The so-called 'aggregated' models are described by a relatively small number of compartments. Usually for models of this type many results about their dynamic properties can be obtained analytically. Examples of 'aggregated' models are the model of Logofet and Alexandrov (1984), Kwon and Schnoor (1994), Mukherjee et al. (2002), Svirezhev and von Bloh (1997), Svirezhev et al. (1999), Svirezhev (2002), Eliseev and Mokhov (2007). Other models are based on a detailed description of both physiological and biochemical processes determining the vegetation productivity and also environmental conditions in which these processes are functioning. The models of this category require that a large number of quantitative empirically determined characteristics of the processes under study are known. These models are sensitive with respect to the choice of the governing parameters for vegetation structure. These parameters are known with a very low accuracy (Svirezhev, 2002). Analytical analysis is not available for models of such type. Typical models of this set are the model of Foley et al. (1996), Friend et al. (1997), Ito and Oikawa (2002), Komarov et al. (2003), Sitch et al. (2003), Nalder and Wein (2006).

In the present paper, we suggest a terrestrial carbon cycle model based on a scheme of the phytomass change, which is continuous in time. This approach is a generalization of that employed in conventional compartmental carbon cycle

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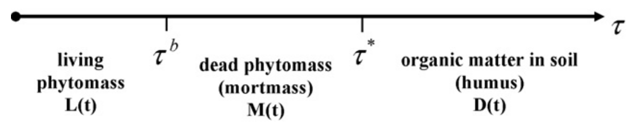

Fig. 1 – The general scheme of the model: phytomass change along the age axis τ.

models. In the latter models, organic matter moves from one compartment to another during its life cycle (say, from living matter to the dead one). In our model, this change is figured via change of parameter (denoted as τ) describing age of phytomass and, in turn, its properties. Parameter τ serves as a 'demographic' marker for phytomass. To the best of the author's knowledge, this approach has not been followed earlier in formulating carbon cycle models. An employment of this approach is a novel feature of the presented model. This method permits us to develop a simple model of carbon exchange for the terrestrial territory under study.

2. Description of model

We consider the following phytomass components in carbon units: photosynthetic (green) phytomass, perennial phytomass (stems, branches, etc.), and root (underground) phytomass. Let us denote by $B_i(t,\tau)$ the phytomass in carbon units of the ith component ($i=1, 2, 3$) for τ "age" at time moment t.

We consider phytomass $B_i(t,\tau)$ along the axis τ (Fig. 1). Initially, we have the living phytomass. Then, at time moment τ_i^b, living phytomass becomes dead and we have the dead phytomass (mortmass). The mortmass is decaying under biological and non-biological processes. At time moment τ_i^*, the mortmass is transformed into organic matter in soil (humus). The humus is slowly decaying organic substance and its lifetime is very large.

According to this scheme, we can write the following equations for the living phytomass, mortmass, and humus:

$$L(t) = \sum_{i=1}^{3} \int_{0}^{\tau_i^b} B_i(t,\tau)\,d\tau, \tag{1}$$

$$M(t) = \sum_{i=1}^{3} \int_{\tau_i^b}^{\tau_i^*} B_i(t,\tau)\,d\tau, \tag{2}$$

$$D(t) = \sum_{i=1}^{3} \int_{\tau_i^*}^{\infty} B_i(t,\tau)\,d\tau. \tag{3}$$

The mass conservation law for the phytomass $B_i(t,\tau)$ takes on the following form:

$$\frac{\partial B_i}{\partial t} + \frac{\partial B_i}{\partial \tau} = -d_i(\tau)B_i - q(\tau)B_i, \tag{4}$$

where $d_i(\tau)$ are the coefficients of decay and $q(\tau)$ is the coefficient of abiotic loss:

$$d_i(\tau) = \begin{cases} 0, & \text{if } \tau < \tau_i^b \\ \mu_i, & \text{if } \tau_i^b \le \tau < \tau_i^*, \\ \eta, & \text{if } \tau \ge \tau_i^* \end{cases} \tag{5}$$

$$d_i(\tau) = \begin{cases} 0, & \text{if } \tau < \tau_i^b \\ q, & \text{if } \tau \ge \tau_i^b \end{cases} \tag{6}$$

We assume that a fraction of the dead organic matter for any age $\tau \ge \tau_i^b$ is lost due to non-biological processes. Boundary conditions for the conservation law have the following form:

$$B_i(t,0) = p_i P(t), \tag{7}$$

where p_i is fraction of ith phytomass component ($i=1, 2, 3$), $P(t)$ is annual net primary production (or, productivity) at some given time t.

According to Eqs. (4)–(7), the phytomass $B_i(t,\tau)$ can be represented as follows:

$$B_i(t,\tau) = \begin{cases} p_i P(t-\tau), & \text{if } 0 \le \tau < \tau_i^b \\ p_i P(t-\tau)e^{-\mu_i'(\tau-\tau_i^b)}, & \text{if } \tau_i^b \le \tau < \tau_i^* \\ p_i P(t-\tau)e^{-\mu_i'(\tau_i^*-\tau_i^b)}e^{-\eta'(\tau-\tau_i^*)}, & \text{if } \tau_i^* \le \tau \end{cases} \tag{8}$$

where $\mu_i' = \mu_i + q$ and $\eta' = \eta + q$.

Taking into account Eq. (8), Eqs. (1)–(3) are rewritten in the following forms:

$$L(t) = \sum_{i=1}^{3} p_i \int_{0}^{\tau_i^b} P(t-\tau)\,d\tau, \tag{9}$$

$$M(t) = \sum_{i=1}^{3} p_i \int_{\tau_i^b}^{\tau_i^*} P(t-\tau)e^{-\mu_i'(\tau-\tau_i^b)}\,d\tau, \tag{10}$$

$$D(t) = \sum_{i=1}^{3} p_i e^{-\mu_i'(\tau_i^*-\tau_i^b)} \int_{\tau_i^*}^{\infty} P(t-\tau)e^{-\eta'(\tau-\tau_i^*)}\,d\tau, \tag{11}$$

If the values of τ_i^b, τ_i^*, μ_i, and time dynamics $P(t)$ are known, then we can calculate the magnitudes of $L(t)$, $M(t)$, and $D(t)$.

Productivity of terrestrial ecosystems $P(t)$ depends on climatic and environmental conditions, in particular on the total amount of atmospheric carbon $C(t)$. In reality, the atmospheric carbon concentration continues to rise (Solomon et al., 2007) and function $C(t)$ is characterized by an exponential dependence on time (Krapivin et al., 1982; Tarko, 2005). This argument allows us to assume the following temporal curve for productivity:

$$P(t) = \begin{cases} P_0, & \text{if } t < t_0, \\ P_0 e^{\lambda(t-t_0)}, & \text{if } t \ge t_0 \end{cases} \tag{12}$$

where λ is parameter determining impact of the carbon concentration in the atmosphere on vegetation, t_0 is the beginning

Table 1 – Specific carbon concentrations (%) for different ecosystems

Ecosystem	Fractions of organic matter			
	Phytomass	Mortmass	Organic matter in soil and peat in bog	Peat in fen
Tundra	56	61	58	56
Taiga forest	40	45	58	56
Broadleaved forest	53	56	58	56
Meadow steppe	37	42	58	–
True steppe	38	43	58	–
Semi-desert	45	47	58	–

of the industrial era. It should be noted that a response of the different vegetation patterns, biomasses, and ecosystems to increase of the carbon in the atmosphere differs from each other.

Thus, we have a closed system of equations for the description of carbon dynamics in terrestrial ecosystems.

3. Calibration: method, material, and discussion

The strong assumption that any local carbon cycle is in a steady state – the so-called steady-state hypothesis – was used in calibrating many existing dynamic models of the global carbon cycle. In other words, it was assumed that, before some moment, which is referred to as a beginning of the industrial era, all local carbon cycles were balanced. For example, this hypothesis was used in calibrating spatial patterns of the global carbon cycle created within the framework of such models as the Moscow Global Biosphere Model (Krapivin et al., 1982; Svirezhev et al., 1985), the Frankfurt Biosphere Model (Ludeke et al., 1994), the Vegetation Continuous Description Model (Brovkin et al., 2002), the Lund–Potsdam–Jena Dynamic Vegetation Model (Sitch et al., 2003).

In our model, in contrast, the steady-state hypothesis is not employed. The model is applied for the European territory of Russia (ETR) as a case study. The European part of Russia is a vast territory with area approximately 3.7×10^6 km^2 that is located between 30° and 60° eastern longitude and 45° and 70° northern latitude. We consider six ecosystems on this territory: tundra, taiga forest, broadleaved forest, meadow steppe, true steppe, and semi-desert. The proportions between the areas occupied by these ecosystems are given in Fig. 2. It should be noted that forest ecosystems occupy more than one-half of the territory under consideration.

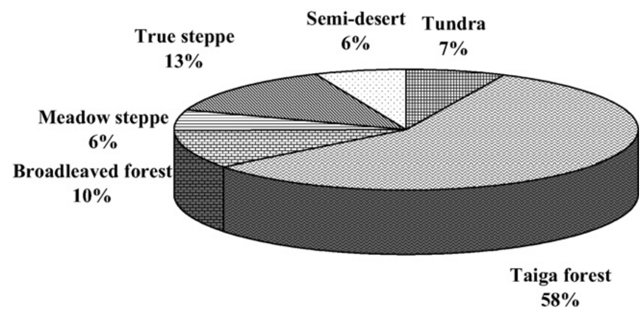

Fig. 2 – Proportion of areas occupied by ecosystems in European Russia.

In order to calibrate the model, we have used extensive experimental information about annual net primary production, annual litter fall, density of the living phytomass and mortmass (Bazilevich, 1993; Bazilevich et al., 1986; Bobkova and Galenko, 2001; Tishkov, 2005; Usoltsev, 2001, 2007), and density of the humus (Orlov et al., 1996; Orlov, 1999). Using specific carbon concentrations for different ecosystems (Golubyatnikov et al., 1998; Orlov et al., 1996; Titlyanova et al., 2005) (Table 1) we have expressed empirical data in the carbon units (Table 2).

Among the ecosystems under consideration, the highest living phytomass per unit area are concentrated in broadleaved forest. The value of the living phytomass decreases sharply to the south and decreases more gradually to north of it. The ecosystems of ETR contain more than 27 Gt of carbon in living phytomass. Meadow steppe ecosystem has the maximum value of the productivity per unit area. The productivity smoothly decreases to the south and north of meadow steppe ecosystem. It is noted that for tundra, taiga forest, and broadleaved forest ecosystems, the aboveground living phytomass and productivity are higher than the

Table 2 – Characteristics for ecosystems of the European territory of Russia

Ecosystem	Living phytomass		Net primary production		Mortmass		Litter fall		Humus	
	kg C/m^2	Gt C	kg C/m^2 per year	Gt C per year	kg C/m^2	Gt C	kg C/m^2 per year	Gt C per year	kg C/m^2	Gt C
Tundra	0.84	0.21	0.11	0.03	2.82	0.71	0.10	0.03	15.99	3.99
Taiga forest	9.17	19.45	0.34	0.72	2.61	5.54	0.24	0.52	26.18	55.49
Broadleaved forest	18.28	6.76	0.64	0.24	2.28	0.84	0.40	0.15	23.63	8.74
Meadow steppe	1.71	0.36	0.71	0.15	1.09	0.23	0.49	0.10	26.44	5.55
True steppe	0.45	0.22	0.48	0.24	0.49	0.24	0.32	0.16	20.29	9.94
Semi-desert	0.58	0.13	0.33	0.08	0.40	0.09	0.26	0.06	6.52	1.50

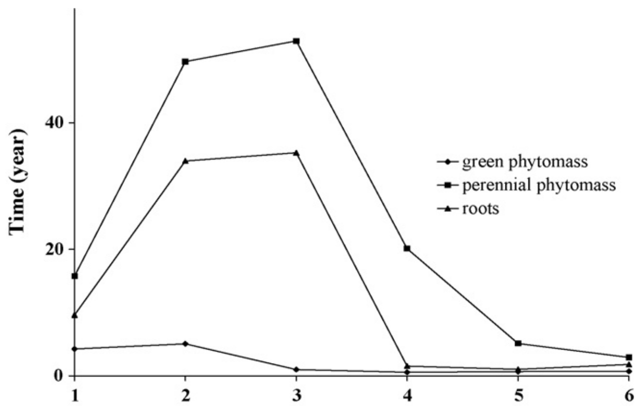

Fig. 3 – Empirical estimations of the ages τ^b for various phytomass components of the ecosystems of the European territory of Russia: 1, tundra; 2, taiga forest; 3, broadleaved forest; 4, meadow steppe; 5, true steppe; 6, semi-desert.

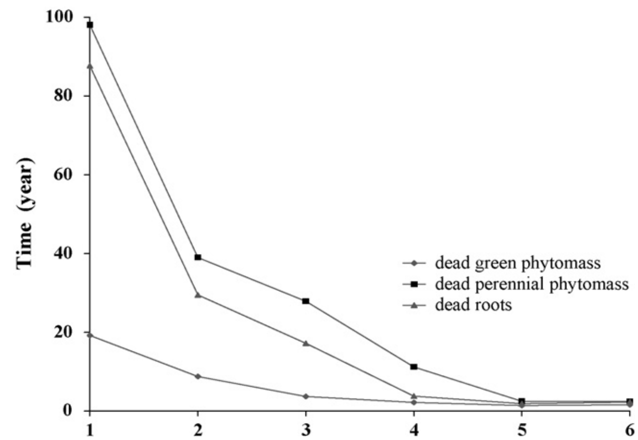

Fig. 4 – Model estimations of mortmass decay time for ecosystems of the European territory of Russia. Notations are the same as for Fig. 3.

underground one. For Steppe and semi-desert ecosystems, the underground living phytomass and productivity are higher than the aboveground one. The annual productivity of the ecosystem characterizes the annual carbon fluxes from the atmosphere into phytomass for ecosystem. For ecosystems of ETR we have obtained the following estimation for this flux: every year tundra ecosystem absorbs about 0.03 Gt of carbon, taiga ecosystem absorbs about 0.7 Gt of carbon, meadow steppe ecosystem absorbs about 0.15 Gt of carbon, true steppe and broadleaved forest ecosystems absorb about 0.2 Gt of carbon, and semi-desert ecosystems absorb smaller than 0.08 Gt of carbon. According to experimental data, the total annual carbon flux from the atmosphere into living phytomass for ecosystems of ETR is 1.4 Gt. Note that simulated estimations of the flux for this region vary between 0.9 and 1.8 Gt carbon per year (Golubyatnikov and Denisenko, 2001).

Part of the living phytomass becomes dead and enters to mortmass storage. The lowest annual litter fall per unit area is registered in tundra ecosystem (about 100 g C/m^2 per year), and the highest value is noted in meadow steppe ecosystem (about 490 g C/m^2 per year). The annual carbon flux from the living phytomass into mortmass is about 1 Gt. The mortmass reaches highest values per unit area in tundra ecosystem where lowest destruction rates of dead organic matter are stipulated by low temperature. The value of the mortmass per unit area smoothly decreases from northern ecosystems to the southern one. For ecosystems of ETR, we have estimated the carbon content in mortmass as high as 7.7 Gt.

According to Orlov et al. (1996), the carbon storage in the uppermost meter of soil of ETR is more than 85 Gt. Soils of taiga and meadow steppe ecosystems are characterized by the higher values of the carbon storage (about 26 kg/m^2). Semi-desert soils have low level of carbon content (about 7 kg/m^2).

Using the conditions $\tau_i^b = B_i/P_i$ we can calculate the ages τ_i^b for various phytomass components of the ecosystems (Fig. 3). It should be noted that the ages τ_i^b are the residence times for carbon in the fractions of living phytomass. The maximal carbon residence times are characteristic for forest ecosystems of ETR where the carbon storage in the living perennial phytomass has the highest values.

Following (Berg and Agren, 1984; Grishina et al., 1990; Kobak, 1988; Semenov et al., 2004), we assume that mortmass decomposition rate is characterized by an exponential curve in time. From this hypothesis and empirical data on mortmass and litter fall (Bazilevich, 1993; Bazilevich et al., 1986), we can estimate the coefficients of decay $d_i(\tau)$ and the respective time τ_i^{dec} for ecosystems on the territory under consideration.

We have obtained the estimates of mortmass decay time τ_i^{dec} for ecosystems of ETR (Fig. 4). The maximal time is characteristic for tundra ecosystem, where low temperature does not permit micro-organisms to decay organic matter. The minimal time of mortmass decay is characteristic for true steppe ecosystem, where climatic conditions and chemical composition of mortmass are optimal for destructive processes. It is noted that the estimates of mortmass decay time for forest and steppe ecosystems are known from the field experiments of other authors (Bazilevich, 1993; Bobkova and Galenko, 2001; Grishina et al., 1990; Storozhenko, 2000). According to those experimental data, time of decay for dead photosynthetic and perennial phytomass are 3–8 and 20–45 years, respectively, in forest ecosystems and 1–2 and 3–15 years, respectively, in steppe ecosystems. Our model estimates of mortmass decay time are 3.7–8.8 and 1.4–2.2 years for dead photosynthetic phytomass of forest and steppe ecosystems, respectively, and 27.9–39.0 and 2.5–11.2 years for perennial phytomass of the same ecosystems, respectively. Thus, our estimates of these time markers coincide closely much with the field experiments results.

If time scales τ_i^b and τ_i^{dec} are known, we can estimate ages τ_i^*, $\tau_i^* = \tau_i^b + \tau_i^{dec}$. Thus, we have two Eqs. (10) and (11) in two unknowns λ and q. Using empirical data on the magnitudes of mortmass, humus, and productivity we can find these parameters as well.

According to our computations, the deviations of the model result from its observational values for living phytomass, mortmass, and humus no more than 2–4%.

4. Estimation of the total exchange carbon flux for ecosystems of ETR

The suggested model permits us to characterize terrestrial ecosystems as carbon sources or carbon sinks and to evaluate intensity of these sources and sinks. Earlier (Golubyatnikov et al., 1998) we approached this problem by modelling the total exchange carbon flux for terrestrial ecosystems. We defined this flux as the difference between carbon income into the net primary production and its emission from dead organic matter (mortmass and humus) due to both biological and non-biological processes. According to this definition and our carbon cycle model, we can represent the total exchange flux of carbon by:

$$W(t) = P(t) - \sum_{i=1}^{3} \mu_i p_i \int_{\tau_i^b}^{\tau_i^*} P(t-\tau) e^{-\mu_i'(\tau-\tau_i^b)} \, d\tau - \eta \sum_{i=1}^{3} p_i e^{-\mu_i'(\tau_i^*-\tau_i^b)} \int_{\tau_i^*}^{\infty} P(t-\tau) e^{-\eta'(\tau-\tau_i^*)} \, d\tau. \qquad (13)$$

From Eqs. (12) and (13) we get an analytic solutions for $W(t)$. For instance, if $\tau_i^b < t - t_0 < \tau_i^*$ then the total exchange carbon flux is as follows:

$$W(t) = P(t)\left(1 - \sum_{i=1}^{3} \mu_i p_i \left[\frac{1}{\lambda+\mu_i'} e^{-\lambda\tau_i^b} + \frac{\lambda}{\mu_i'(\lambda+\mu_i')} e^{\mu_i'\tau_i^b-(\mu_i'+\lambda)(t-t_0)} - \frac{1}{\mu_i'} e^{-\mu_i'(\tau_i^*-\tau_i^b)-\lambda(t-t_0)}\right] - \frac{\eta}{\eta'} e^{-\lambda(t-t_0)} \sum_{i=1}^{3} p_i e^{-\mu_i'(\tau_i^*-\tau_i^b)}\right) \qquad (14)$$

If value of $W(t)$ at a particular time t for a given ecosystem is positive, then carbon release into the atmosphere is smaller than its absorption and an ecosystem serves as sink of carbon. In contrast, if value of $W(t)$ at a particular time t for a given ecosystem is negative, then carbon release into the atmosphere is higher than its absorption and this ecosystem is a source of carbon.

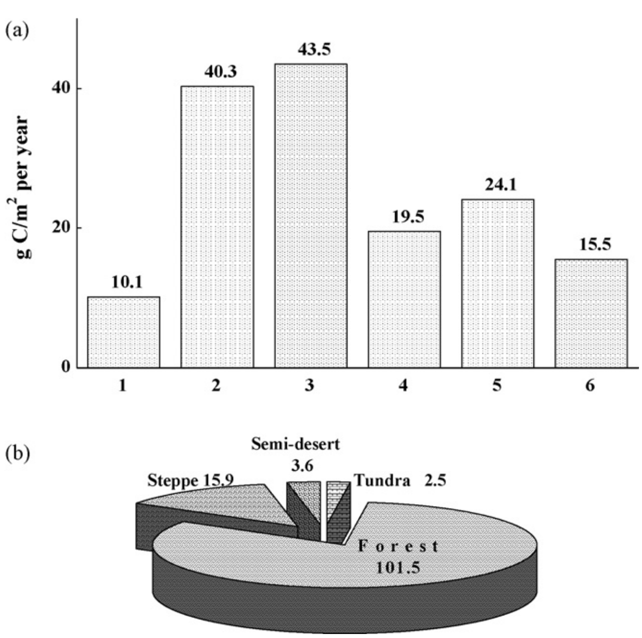

Fig. 5 – Estimations of annual total exchange carbon flux for ecosystems of the European territory of Russia ((a), g C/m^2 per year; (b), Mt C per year). Notations are the same as for Fig. 3.

Using the experimental data over the second half of the 20th century, we have calculated the annual total exchange carbon flux for ETR ecosystems (Fig. 5). The maximum magnitude of this carbon flux is characteristic for broadleaved forest (about 44 g C/m^2 per year); the minimum magnitude is characteristic for tundra (about 10 g C/m^2 per year).

We have found that the territory of European Russia between 55° and 60° northern latitude exhibits the high magnitudes of the annual total exchange of carbon. This territory is occupied by forest phytocenoses of southern taiga and broadleaved forests. We have shown that a significant role of carbon accumulators belongs to forest ecosystems of ETR, which deposit about 100 Mt carbon per year. This value is in agreement with estimates obtained by other authors for forest ecosystems (Isaev and Korovin, 1999; Zamolodchikov et al., 2005; Zavarzin, 2007). According to our calculations, steppe ecosystem absorbs up to 16 Mt carbon per year. In semi-desert ecosystem, annual carbon sink magnitude decreases to 4 Mt. Tundra ecosystem is characterized by the minimal difference between carbon absorption and its emission into the atmosphere, about 2.5 Mt per year. The annual total exchange carbon flux for ETR is evaluated as 124 Mt C. Thus, at the end of the last century for the European territory of Russia carbon release into the atmosphere was smaller than its absorption and, hence, this territory was a carbon sink. The obtained estimates are in agreement with those of other authors (Mokronosov and Kudeyarov, 1997; Zavarzin, 2001, 2007 and other) that European Russia is a sink of carbon.

5. Conclusion

A carbon cycle model is presented which allows us to get an analytic solution and does not employ the steady-state hypothesis for preindustrial local carbon cycle. The model is based on a scheme of phytomass change, which is continuous in time. It is calibrated using extensive experimental information on living and dead phytomass, soil organic matter, plant productivity, and litter fall. In our model, organic matter stock changes are described via changes of phytomass age. This approach is a novel feature of the model.

The proposed model is a new tool to describe and understand complex interactions in terrestrial ecosystems. It enables us to study of carbon dynamics in different pools of terrestrial ecosystems under the climate change and to estimate the respective carbon budget.

The suggested model permits us to characterize a particular terrestrial territory as a source or a sink for carbon. Our research has shown that the European territory of Russia was a carbon sink at the end of the 20th century. We have evaluated the intensity of this sink as 124 Mt of carbon per year. We have found that a significant role of carbon accumulators belongs to forest ecosystems of European Russia. These ecosystems accumulate about 100 Mt carbon per year.

Certainly, the model cannot take into account the whole set of productive and destructive processes for organic matter. Therefore, the results obtained represent some estimates but they nevertheless give an idea of the qualitative pattern of the carbon balance for the territory under study.

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